Result 1 (reindexed basis vector). [mathlib, Module.Basis.reindex_apply]
Let K, I and J be types. Let *field* be a field structure on K. Let M be a module over K. Let b be a basis of M indexed by I. Let e be an equivalence from I to J. Let j be an element of J. Then

$$b[e]_j = b_{e^{-1}(j)}.$$

Adapted from the cited source.

Result 2 (equivalence backward forward). [mathlib, Equiv.symm_apply_apply]
Let A and B be types. Let e be an equivalence from A to B. Let x be an element of A. Then

$$e^{-1}(e(x)) = x.$$

Adapted from the cited source.

Result 3 (equivalence forward backward). [mathlib, Equiv.apply_symm_apply]
Let A and B be types. Let e be an equivalence from A to B. Let y be an element of B. Then

$$e(e^{-1}(y)) = y.$$

Adapted from the cited source.

Result 4 (identity equivalence fixes points). [mathlib, Equiv.refl_apply]
Let A be a type. Let x be an element of A. Then

$$\mathrm{id}_A(x) = x.$$

Adapted from the cited source.

LLM-assisted statements and proofs (gpt-6, 2026-11-07). See PROVENANCE.md. Reindexing uses an equivalence from old indices to new indices. The surviving-label equivalence for interval-basis restriction remains to be constructed.